

Athanasios (Nassos) Moschos

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<https://0ena.github.io>

SHORT SKILL SET AND RESEARCH INTRODUCTION

My research interests lie at the intersection of hardware and systems security, building on my background in IC design, computer architecture and side-channel analysis. For my research, I implement RISC-V microarchitectures as either tape-out-ready ASIC blocks or FPGA prototypes to study their susceptibility to insertion of malicious functionality, commonly referred to as hardware trojans. Through this, I examine the system-level manifestations of trojan activity during offensive campaigns to identify methods for detecting and mitigating such attacks. My research has given me a strong understanding of the cross-layer interactions between the processor hardware and software, enabling me to perform RLT-level microarchitectural modifications, while preserving performance and native design functionality. To evaluate the physical impact of RTL modifications, I developed a fully automated RTL-to-GDSII flow based on industry EDA tools. I have experience with C/C++, Python and Matlab, which I have employed in both industry and academia. During my Master's, I have completed relevant coursework in computer architecture and memory technologies, developing C/C++ simulators for cache memories, branch predictors, FOCO architecture (Tomasulo's algorithm) and cache coherence protocols. I also use Sed/AWK/Python for processing and analyzing large system and design artifacts (*i.e.*, memory traces, ASIC netlists). Lastly, I maintain my own Linux-based EDA-tool infrastructure (Cadence, Synonpsys, Mentor) required for my work.

EDUCATION

Ph.D. in Electrical and Computer Engineering January 2020 - December 2026 (**Expected**)
Georgia Institute of Technology, [COEUS Center](#) Atlanta, GA
Advisor: [Dr. Angelos Keromytis](#)

M.S. in Electrical and Computer Engineering June 2024
Georgia Institute of Technology Atlanta, GA

Diploma in Electrical and Computer Engineering (B.Eng. & M.Eng.) July 2017
University of Patras, [VLSI Design Laboratory](#) Patras, Greece
Advisors: Dr. Odysseas Koufopavlou and [Dr. Apostolos Fournaris](#)

SKILL SET

EDA Tools: Vivado, Genus/Innovus/Tempus, VCS, Questasim, Virtuoso, LEC, Spectre
Programming: System Verilog, VHDL, Python, C/C++, Matlab, TCL, Assembly, Sed/Awk, Bazel
Engineering: FPGA prototyping, measurements with oscilloscopes, SDRs and EM probes. Linux servers & EDA tools setup and management.
Languages: English, German, Greek.

INDUSTRY EXPERIENCE

CPU Design Verification Engineer - RISC-V Team May 2024 - December 2024
[Tenstorrent](#) Austin, Texas

- Identified simulation bottlenecks caused by the RTL or the design verification logic in RISC-V HPC uArchs, using VCS profiler and Linux Perf for Verilator. Performed optimizations that led to significant reductions in simulation time (up to 14%), across different test-benches and uArch modules, thus lowering the simulation licensing costs. Automated through scripting the discovery of commits in the RTL codebase that throttle simulation performance.
- Reduced ZeBu server's resource utilization to increase emulation capacity. In particular, I performed optimizations in the RTL and C++ API of the Performance Monitor Unit (PMU) used in HPC uArchs, that led to a 40% reduction in the utilization of FPGA resources related to waveform capturing. The optimizations enable dynamic reconfiguration of the PMU's performance counter widths, depending on the needs of the verification team.
- Worked on establishing the infrastructure for gate-level simulations of RISC-V HPC uArchs. Reconfigured and enhanced the existing Bazel-based automation framework, to strip the test-bench platform from DV related logic (e.g., XMRs) and allow execution of verification testing on the bare DUT. The enhanced features provided plug-n-play capabilities, where the RTL logic of the DUT could be swapped with the DUT's gate-level netlist.

Mixed-Signal/Digital IC Design Engineer
weasic Microelectronics

April 2018 - December 2019
Athens, Greece

- Participation in **5 first pass silicon success** tape-outs of RF/mixed-signal chips as a Digital IC Design Engineer and Analog Layout Engineer.
- Established the Company's first RTL-to-GDSII digital design flow and implemented the front-end and back-end of various digital blocks (including a 32-bit RISC-V microarchitecture) used in the Company's products .
- Performed layout in different technology processes (55nm SiGe BiCMOS, 45nm GF RF-SOI, 22nm GF FD-SOI) on top-chips, individual analog blocks and custom cells used in the digital design flow. My ability in solving fast DRC and LVS errors in complex top-chip/analog layouts enabled the Company to meet all of its tight tape-out deadlines.
- Digital front-end/back-end experience: RTL coding, design synthesis, floorplanning, power planning, place and route, clock tree synthesis, static timing analysis (STA), sign-off characterization, physical verification, equivalence checking, ECOs.
- Created Sed and Awk scripts to adjust the gate-level netlists of mixed-signal top-chip blocks, to fit the client's digital-on-top flow for functional testing/simulations.

Embedded Systems Software Engineer
TELETEL S.A.

November 2017 - April 2018
Athens, Greece

- Co-leading a small team instructed with the implementation of a testing-suite for the validation of the Xtratum hypervisor on the LEON3 microprocessor. The project was funded by the European Space Agency [\[media\]](#).
- Implementation of software modules in C and C++ that facilitated the communication between the XtratuM hypervisor and a custom-made Python testing-suite.
- Acquired experience in the implementation of uni-tests for the validation of software modules running on the SPARC-V8 architecture.

**RESEARCH
EXPERIENCE**

Graduate Research Assistant
Georgia Institute of Technology

January 2020 - Present
Atlanta, GA

During my PhD research, I investigated hardware trojan attacks on general-purpose computing platforms from a systems-level perspective. Using Linux-capable, 64-bit RISC-V microarchitectures (*i.e.*, CVA6 and Rocket Core) as testbeds, I designed and evaluated hardware trojans that support offensive capabilities against the host system, including code injection, unauthorized access of privileged memory and denial of service. Using FPGA prototyping, I examined the interaction of hardware trojans with the operating system and the processor's software stack (*i.e.*, runtime engines) to evaluate the effectiveness of such attacks. Additionally, I collected and analyzed electromagnetic measurements from the processor's operation to assess the stealthiness of different trojan implementations to side-channel detection. Using TSMC's 28nm HPC+ process, I implemented the digital front-end and back-end of different RISC-V microarchitectures. With the resulting sign-off layouts I studied the susceptibility of such complex ICs to hardware trojan insertion during fabrication.

Prior to the above, I investigated the susceptibility of Intel's AES-NI x86 instruction-set extension to physical side-channel attacks, and used correlation power analysis (CPA) to recover cryptographic keys. For that purpose, I developed signal-processing scripts for pattern recognition, extraction and alignment, filtering, denoising and rejection of low-quality measurements.

Ultimately, my research focuses on exposing cross-layer vulnerabilities in modern processors and developing practical defenses that span the silicon lifecycle, from design and fabrication to deployment.

Physical Design Lead
[Silicon Jackets](#) at Georgia Institute of Technology

January 2025 - September 2025
Atlanta, GA

- Educating a team of 35 Georgia Tech master's and undergraduate students in physical design tools and providing them with a comprehensive understanding of the RTL-to-GDSII flow.
- Led the synthesis and physical implementation workstream for Buzzcore, a 6-stage base-ISA RISC-V core with 4 KB of DFFRAM. Organized and assigned tasks across a team of master's students to keep the design on track for tape-out, while providing guidance on RTL synthesis, design constraints, floorplanning, STA, sign-off characterization, top-chip integration, and physical verification (LVS & DRC) using analog design tools as part of the digital implementation flow. The chip was taped out through Cadence's SKY130 MPW program.

- Mentored two outstanding Master’s students:
 Shibao Su - post-graduation: SoC Physical Design Engineer at Apple
 Matthew Bowlby - post-graduation: Embedded Software Engineer at Garmin

Diploma Thesis Researcher
 University of Patras

June 2015 - April 2017
 Patras, Greece

I studied side-channel attacks (e.g. CPA) and cryptanalysis methods to evaluate the protections of hardware security modules. I designed and implemented a digital controller IP (FPGA prototyping on Xilinx/AMD Spartan-6 chips) that enables fast collection of electromagnetic power measurements from different implementations of cryptographic modules inside FPGAs. The digital controller is parametric, allowing reconfiguration at compile time to accommodate different cryptographic modules. Using the controller implementation, I collected measurements to evaluate the side-channel resistance of several cryptographic modules. Part of my Diploma Thesis led to 3 publications.

TEACHING

Graduate Teaching Assistant, Georgia Institute of Technology

ECE 4115: Introduction to Computer Security

Spring & Fall 2021

Preparing the lab exercises, grading and holding students’ office hours.

HONORS & AWARDS

1st Place in Tenstorrent’s AI Hackathon

August 2024

Built an LLM-based prototype of a Q&A assistant for the engineering teams, deployable on TT’s cards. Each team-member won a [Wormhole n150 card](#).

Young DAC Fellow fellowship from the 61st DAC Conference

June 2024

CSAW 2023 Competition Finalist - AI Hardware Attack Challenge

November 2023

Advanced in the final round of the competition with an AI-generated hardware trojan design crafted for a Linux-capable RISC-V microarchitecture.

Young DAC Fellow fellowship from the 60th DAC Conference

July 2023

Acknowledgment from the 32nd EUROCRYPT Conference

May 2013

One of the team members helping organizing Eurocrypt 2013 Conference in Athens, Greece.

Travel Grants

DAC 2023/2024, CSAW 2023, CHES 2022, EUROCRYPT 2014/2015

PUBLICATIONS

1. [A. Moschos](#), B. B. Yilmaz, K. Valakuzhy, and A. D. Keromytis, “Assessing Runtime Electromagnetic Detection of CPU Hardware Trojans Targeting Kernel Memory,” 2026 IEEE International Conference on Electronics Circuits and Systems (ICECS), 2026.
2. [A. Moschos](#), F. N. Monroe, A. D. Keromytis, “Towards Practical Fabrication Stage Attacks Using Interrupt-Resilient Hardware Trojans”, 2024 IEEE International Symposium on Hardware Oriented Security and Trust (HOST), 2024, [\[media1\]](#)
3. G. Kokolakis, [A. Moschos](#), A. D. Keromytis, “Harnessing the Power of LLMs in Hardware Trojan Design”, 2024 Artificial Intelligence in Hardware Security, Applied Cryptography and Network Security Workshops (ACNS), 2024.
4. [A. Moschos](#), K. Valakuzhy, A. D. Keromytis, “On the Feasibility of Remotely Triggered Automotive Hardware Trojans”, 2022 International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME), 2022, [\[media1\]](#) [\[media2\]](#).
5. A. Fournaris, [A. Moschos](#), N. Sklavos, “Side Channel Assessment Platforms and Tools for Ubiquitous Systems”, Book: Security of Ubiquitous Computing Systems, 2021, Springer.
6. A. Fournaris, Charalambos Dimopoulos, [A. Moschos](#), O. Koufopavlou. “Design and leakage assessment of side channel attack resistant binary edwards Elliptic Curve digital signature algorithm architectures”, Microprocessors and Microsystems 64: 73-87, 2019.
7. [A. Moschos](#), A. Fournaris, O. Koufopavlou, “A flexible leakage trace collection setup for arbitrary cryptographic IP cores”, 2018 IEEE International Symposium on Hardware Oriented Security and Trust (HOST): 138-142, 2018.

POSTERS

1. [A. Moschos](#), A. D. Keromytis, “Considering the Future of Hardware Trojan Attacks”, DAC Young Fellows Poster Session - 60th Design Automation Conference (DAC), 2023.
2. [A. Moschos](#), A. D. Keromytis, “The Design and Implementation of an Open-Source Hardware Trojan for a 64-bit RISC-V CPU Design”, Poster Session - IACR Transactions on Cryptographic Hardware and Embedded Systems (TCHES), 2022.

SERVICE *U.S. Open-Source Software Security Initiative Workshop* August 2022
Scribe for the “[Memory-Safe Language Adoption in OSS](#)” session.

INVITED TALKS *Athecrypt 2018* January 2018
Presentation on “Automated Functional Validation and Security Evaluation Setup for Arbitrary Cryptographic IP cores”.

**ADDITIONAL
SCHOOLS &
TRAINING** *Hardware.io USA 2026* May 2026
Training School on “Integrated Circuits Reverse Engineering: ROM is your primary target” in Santa Clara, USA.

TRUDEVICE 2014 July 2014
Training School on “Trustworthy Manufacturing and Utilization of Secure Devices” in Lisbon, Portugal.

**EXTRA-
CURRICULAR
ACTIVITIES** I hold a skipper’s license since 2015 and I have been involved in yacht racing with large keel boats. I also enjoy diving with friends in my free time, as I have a free-diving certification from the Greek Diving Association and an open water diver certification from PADI.

References

Angelos Keromytis, Professor at Georgia Tech: angelos@gatech.edu - my PhD Advisor
Fabian Monrose, Professor at Georgia Tech: fabian@ece.gatech.edu - close collaborator
Varun Koyyalagunta, Sr. Principal Engineer at Tenstorrent: vkoyyalagunta@tenstorrent.com - my supervisor at TT
Emmanouil Metaxakis, CEO & Co-Founder at Weasic Microelectronics: emetax@weasic.com - my supervisor at Weasic